EXPERIMENTAL LABORATORY REPORT

NiO Electrospun Fiber Networks for Lithography-Free Memristive Junctions

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Year	2023–2026
Work type	Independent work, original concept
Key sample	0301 (solution 03, sample 01; 17 cm, 12–15 kV)

Abstract

Electrospinning was used to build NiO fiber networks as a lithography-free route to memristive junctions. The resulting network on glass showed nA-level, hysteretic and rectifying I–V behavior over ±10 V. Key outcomes:

- **Concept:** spin a NiO fiber layer, then an insulating PVP fiber layer, so each crossing acts as a memristive junction.
- **Process study:** six electrospinning conditions (10–17 cm, 5–15 kV) were mapped by optical microscopy and needle-tip imaging.
- **Key process finding:** with a water-based solution, tip-to-collector distance matters more than field strength. Best window: 15–17 cm at about 12–15 kV.
- **Device result:** pinched hysteresis in both polarities, ≈ 6 nA at +10 V, I_+/I_- ≈ 3, and an NDR-like anomaly near −7 to −6 V.
- **Status:** addressable junctions were not achieved. The inert-gas anneal carbonized the PVP spacer, the dense mat formed a percolating network, and the switching origin is unresolved.

1. Objective and Concept

The goal was a many-junction memristor array without high-resolution lithography, which the lab did not have. The approach was:

1. Electrospin NiO-precursor fibers as the switching material.
2. Electrospin insulating PVP fibers over them.
3. Treat every NiO/insulator crossing as one memristive junction, so junction density is set by fiber density rather than by a mask.

The approach trades lithographic control for speed and cost: one spinning run can create thousands of crossings.

2. Experimental

2.1 Materials and procedure

All steps used standard lab equipment. Quantities that were not recorded are marked as such rather than estimated.

Table 1. Experimental procedure and recorded details.

#	Operation	Description	Recorded details
1	Solution	Nickel nitrate ($Ni(NO_3)_2 \cdot 6H_2O$) + PVP in water, stirred with heating	< 1 h stirring; low viscosity (water-like). Masses not recorded. Solution code: 03
2	Electrospinning (NiO layer)	Standard syringe needle, horizontal needle–collector geometry, Al foil collector, random fibers	≈ 15 min per layer; 10–17 cm, 5–15 kV (see Section 3.1). Flow rate and needle gauge not recorded
3	Spacer layer	PVP fibers electrospun on the same NiO mat, same setup	PVP concentration not recorded
4	Anneal	Tube furnace (CVD system) under Ar or N_2	Temperature, time and ramp not recorded. Sample turned grey
5	Transfer	Calcined mat peeled off the Al foil and placed on a glass slide	Mat stayed intact after peeling
6	Contacts	Two Ag-paste dots on the fiber mat	Contact size and gap not recorded
7	Measurement	Lateral 2-wire I–V sweep, ±10 V	Two sweeps recorded

2.2 Sample designation

Sample 0301 denotes solution 03, sample 01, spun at 17 cm and 12–15 kV.

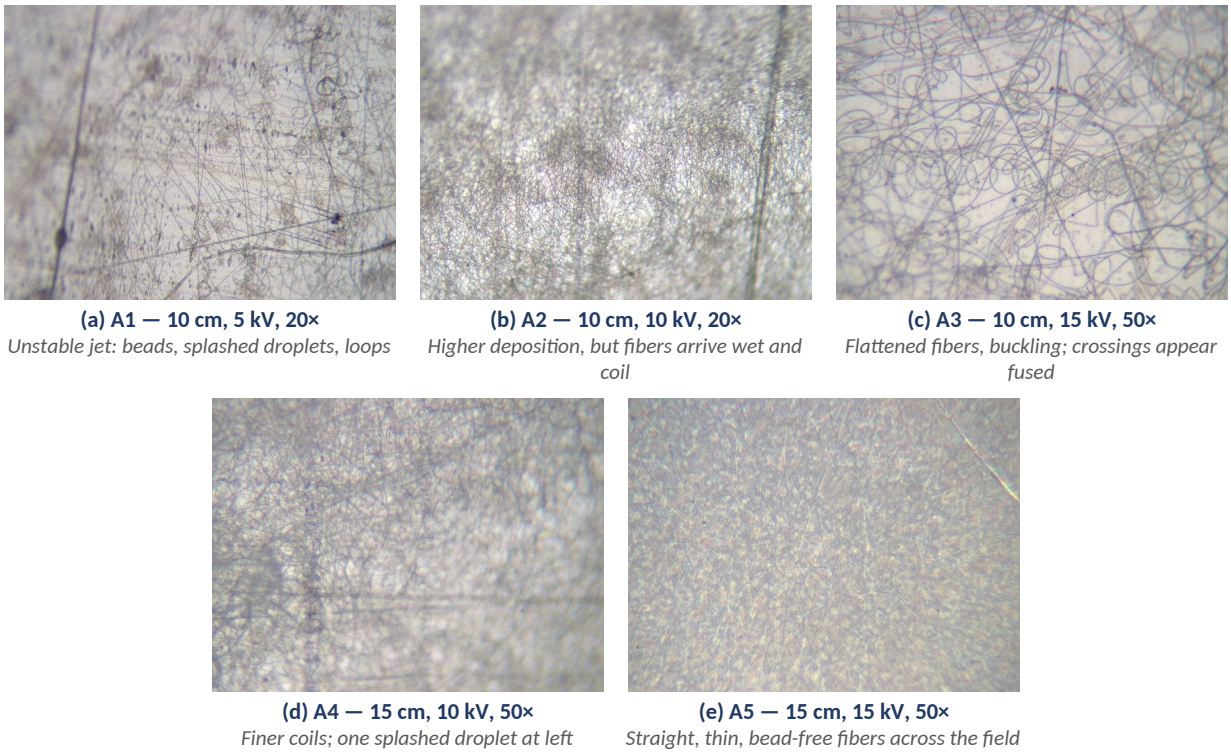
3. Results and Discussion

3.1 Electrospinning parameter study

Distance controlled fiber quality more than field strength: at the same 1.0 kV/cm, 10 cm gave a wet, coiled mat while 15 cm gave thin, straight fibers (Table 2, Figure 1).

Table 2. Electrospinning conditions and observed morphology.

Run	Distance	Voltage	Field (kV/cm)	Objective	Morphology
A1	10 cm	5 kV	0.50	20×	Random fibers, loops, beads, many splashed droplets, uneven diameter
A2	10 cm	10 kV	1.00	20×	Dense, thick mat; heavy coiling; out-of-focus depth
A3	10 cm	15 kV	1.50	50×	Thick, likely ribbon-like fibers; strong buckling; fused crossings
A4	15 cm	10 kV	0.67	50×	Dense mat of finer fibers; smaller coils; one large splash
A5	15 cm	15 kV	1.00	50×	**Thin, straight, uniform fibers; no beads (best)**



(a) A1 — 10 cm, 5 kV, 20×
Unstable jet: beads, splashed droplets, loops

(b) A2 — 10 cm, 10 kV, 20×
Higher deposition, but fibers arrive wet and coil

(c) A3 — 10 cm, 15 kV, 50×
Flattened fibers, buckling; crossings appear fused

(d) A4 — 15 cm, 10 kV, 50×
Finer coils; one splashed droplet at left

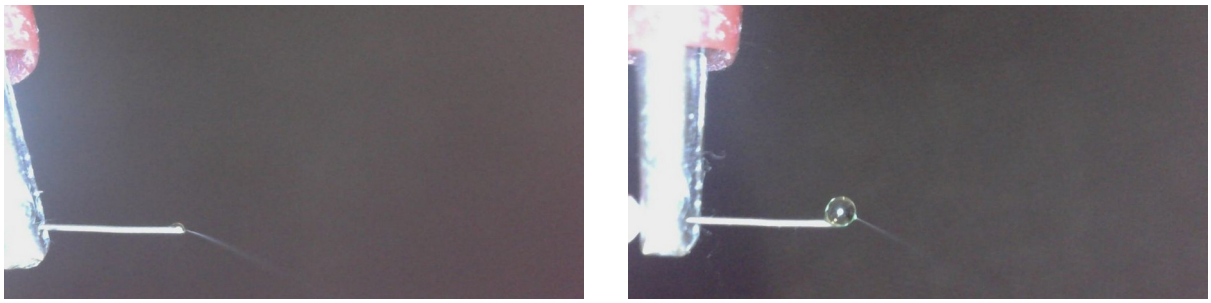
(e) A5 — 15 cm, 15 kV, 50×
Straight, thin, bead-free fibers across the field

Figure 1. Optical micrographs of the electrospun mats for conditions A1–A5. Micrographs carry no scale bar; only the objective magnification is known.

Interpretation. Water evaporates slowly, so flight time decides whether fibers dry before landing. At 10 cm, raising the voltage only increased deposition of wet fibers; at 15 cm, it stretched dry fibers further. Low voltage (5 kV) gave an unstable jet with beads. Because no scale bar is available, no fiber diameters are claimed.

3.2 Jet imaging at the needle tip

At 17 cm with solution 03, the jet was closer to a stable cone-jet at 12 kV than at 15 kV; no Taylor cone formed at 15 kV (Figure 2).



(a) B1 — 17 cm, 12 kV
Small meniscus; jet leaves from the needle apex

(b) B2 — 17 cm, 15 kV
Spherical droplet; jet leaves from its side, off-axis

Figure 2. Needle-tip frames at 17 cm (solution 03), taken from the original process videos: (a) 12 kV, (b) 15 kV.

- **Surface tension:** water (≈ 72 mN/m) resists forming a Taylor cone, so the droplet stays spherical.
- **Flow balance:** a large droplet suggests the feed rate exceeded the jet's take-off rate. Solution accumulates, the jet starts, the droplet drains, and the jet stops. This matches the intermittent jetting noted in the 12 kV recording at higher voltage.

- **Downward jet in both frames:** the deflection does not depend on voltage, so it most likely comes from the setup geometry (needle–collector misalignment or gravity).
- **Link to morphology:** droplet shedding explains the splashes, beads and uneven diameters in A1–A4.

3.3 Electrical characterization

The calcined fiber mat on glass showed pinched hysteresis in both polarities at the nA level, with clear rectification (Figure 3, Table 3).

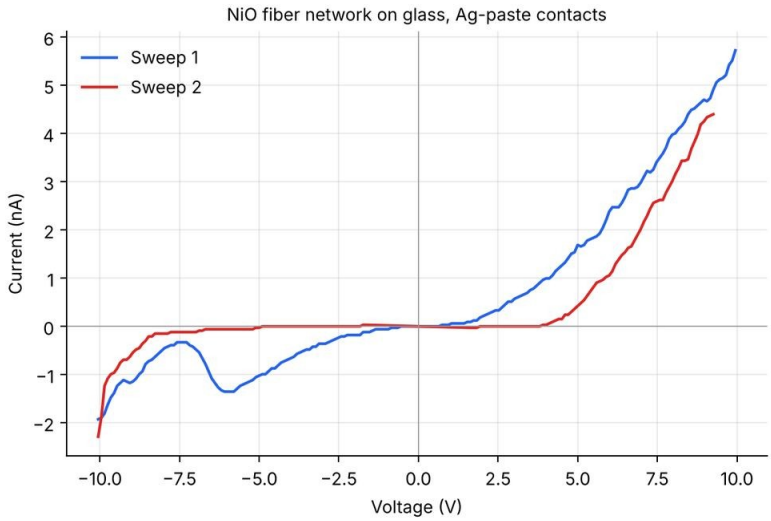


Figure 3. Lateral 2-wire I–V of the NiO fiber network over ±10 V, two sweeps. Ag-paste contacts on the fiber mat, glass substrate.

Table 3. Summary of electrical observations.

Observation	Value
Current at +10 V	≈ 4.4–5.7 nA
Current at −10 V	≈ −2 nA
Rectification I_+/I_- at ±10 V	≈ 3
Resistance scale	≈ 2 GΩ at 10 V
Hysteresis	Sweep 1 above sweep 2 in both polarities
Anomaly	NDR-like dip and peak in sweep 1 between −7.5 and −6 V

4. Critical Assessment

The measured device was a random NiO-based fiber network, not an array of isolated NiO/insulator junctions (Table 4).

Table 4. Intended versus most likely device structure.

Element	Intended	Most likely in practice	Evidence
NiO fibers	Switching layer	NiO–carbon composite, possibly non-stoichiometric	Grey color; anneal in Ar/N_2
PVP fibers	Insulating spacer	Pyrolyzed to carbon residue	No oxygen to burn PVP in an inert anneal
Crossings	Isolated junctions	Percolating network, many contacts per point	Dense, random mats (A2, A4)
Mat integrity	Brittle oxide	Flexible, peeled intact	Carbon binder holds fibers together
Current path	Through junctions	Lateral, through the whole network	Two Ag contacts on glass

Possible origins of the hysteresis, ordered from most to least likely:

1. **Ag migration** from the paste along the porous mat (electrochemical metallization).
2. **Fiber–fiber junctions** acting as distributed switching nodes.
3. **Charge trapping** in non-stoichiometric NiO, consistent with the NDR-like anomaly.
4. **Intrinsic NiO resistive switching.**

The asymmetry between polarities fits two unequal Ag/fiber Schottky contacts. Separating these origins would need a control sample, XRD and SEM, which were not done.

5. Conclusions

- In a water-based solution, tip-to-collector distance had a stronger effect on fiber quality than field strength; the best window was 15–17 cm at about 12–15 kV.
- The calcined NiO fiber network on glass showed nA-level pinched hysteresis, rectification ($I_+/I_- \approx 3$) and an NDR-like anomaly.
- Addressable junctions were not achieved, and the origin of the switching remains unresolved without control samples, XRD and SEM.

6. Lessons and Next Steps

Reframing. Random, self-assembled memristive nanowire networks are studied as physical reservoirs for neuromorphic computing. The traits seen here (nonlinearity, hysteresis, NDR-like dynamics, many interconnected nodes) are the ones a reservoir needs, which links the fiber work to reservoir computing with thermal neuristors.

Redesign for addressable junctions. Each failure point has a direct fix (Table 5).

Table 5. Identified problems and proposed fixes.

Problem	Proposed fix
Unstable jet, no Taylor cone (water)	Ethanol/water co-solvent to lower surface tension; lower feed rate
Wet, coiled fibers	Longer distance (15–17 cm) and drier solvent mix
Random, dense mat	Aligned collection (rotating drum or parallel-gap collector); shorter spinning time
PVP spacer carbonized	Calcine NiO in air first, then add a heat-stable oxide spacer (e.g. sol-gel SiO_2 or Al_2O_3 fibers)
Ag-paste contacts	Evaporated electrodes through a shadow mask
Switching origin unclear	Control samples (spacer only, contacts only), XRD and SEM

Appendix A. Data Inventory and Limitations

Everything shown comes from original lab material.

Table 6. Data inventory.

Material	Status	Used in
Optical micrographs, 5 conditions (20×/50×)	Original	Figure 1 (A1–A5)
Sample 0301 micrographs (17 cm)	Original	Not shown (A5 represents the best regime)
Process videos, 17 cm at 12 and 15 kV	Original	Figure 2 (B1–B2 frames); video link to add
I–V figure	Original	Figure 3
Raw I–V data	Not available	—
XRD, SEM, EDX	Not measured	—
Post-anneal micrographs	Not available	—
Solution masses, flow rate, anneal temperature	Not recorded	—

Outstanding items

- Add the edited electrospinning video link (12 vs 15 kV).
- Request the original I–V figure or raw file from Prof. Mohseni to confirm sweep direction.